

Interactive Digital Art and Design SE608

Project Proposal Year IV



Student Name:	Aleksandra Skarzycka	Student Number:	C00294397
Working Title:	Environmental Showcase		
Description:	Creating an environmental rendered showcase and using a Particle/Visual Effects Editor to bring life to the atmosphere, giving certain aspects their needed particles or visual effects. I'm aiming to make it an outdoor campsite type of setting, with an active fire lit so I could mess with the fire effects and any lighting the fire would emit and affect the surrounding area. My goal is to make it as detailed as my skills will let me. This project is just inspired by online games I play, currently into Halloween: The Game which has released just recently, and I just enjoy seeing the small details that game creators put into their games that do not have any significance in the actual gameplay, so I've felt the need to create my own environment that isn't a game, but to test the waters and actually get into 3D modelling to see if it would be something I would really enjoy making and improving on. I would be 3D modelling the environmental assets using polygonal modelling, using modifiers where appropriate, or by sculpting the landscape using geometry nodes Using shaders and materials using blender's shader editor, and messing with the settings to my liking Creating the particles mentioned by using blender's classic emitter system, the hair particle system, or the modern geometry nodes where appropriate The VFX would be smoke, fire, fluid etc. where appropriate in the environment Lighting to render natural looking light to match the environment, using area lights, point lights, volumetric lighting etc. A bit of animation to either make the environment lively by making the elements move like particles, objects or any effects. Possible addition of organic models with an idle animation for a livelier scene in the showcase. Configuring render settings for the final render of the showcase Using blender's compositor for bloom/glow, any colour grading, depth of field effects, glare etc. Making sure to optimise the showcase by taking note of polygonal count, texture resolution, particle counts, render time, memory usage etc. Procedural generation for anything that can be repeated so it's not all manually placed Would like to focus on using the Blender software mainly but will also likely experiment with Maya. I mainly want to focus on building the environment, placing everything down and making sure I am satisfied with it, and then add in the details needed like materials, textures, particles, any VFX, and everything else that is mentioned etc.		

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Department of

Computing

C A R L O W

Reasons for selecting project:	I want to dive deeper into building 3D environments as it has been something I've always wanted to learn how to do and get better at, so I think this would be great for me to work on as I'd be actively learning in classes as well, with access to online tutorials or any guides made by the platforms I'd be using, in this case Blender. I may also experiment in Maya as well, but I would like to focus on Blender.
Proposed research content:	https://www.researchgate.net/publication/396266519_VFX_in_Blender_for_scientific_visualisation_and_storytelling https://docs.blender.org/manual/en/latest/modeling/index.html https://docs.blender.org/manual/en/latest/physics/index.html https://docs.blender.org/manual/en/latest/index.html https://libsearch.setu.ie/cgi-bin/koha/opac-detail.pl?biblionumber=380445 https://ebookcentral.proquest.com/lib/itcarlow/detail.action?docID=4180376
External links (if applicable):	
Hardware requirements:	Blender System Requirements Maya System Requirements
Software requirements:	Blender System Requirements Maya Software Requirements
Other requirements:	

Date: 21/09/2026

Signed:

A Sharsyken

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Approved/Not Approved:	
Reasons for not approving project:	

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Conditions attached to approving project:	
Approved/Not Approved:	
Name of Supervisor:	
Signed: _____	Date: _____